Exhibit 3. Where Studio sits in Aladdin

Who builds	Client engineers	Code-curious portfolio managers	Data analysts	Excel users building reports
BlackRock's applications	**About 100 applications** such as Explore, Dashboard and Risk Radar		**Aladdin Copilot, inside them** each tool matches an Aladdin API already in production	
Aladdin Studio	**Aladdin Studio:** the developer and data layer, whose parts Exhibit 4 shows			
Shared services and data	**Portfolio management, trading, risk, compliance, operations, accounting** reached through the 11 public API domains			
Hosting	Microsoft Azure	Amazon Web Services		
Outside inputs	The client's own and third-party data, loaded into Data Cloud or through the Data eXchange, and vendor data such as prices and ESG research			

CONTROLS IN THE API SPECIFICATIONS

Access set per API
Each API has its own OAuth scopes.

Compliance by default on orders
Posting an order runs compliance unless the caller turns it off, and returns any violations.

For illustrative purposes only. Every box is named in the sources below. The 11 domains are the packages of BlackRock's public software development kit (SDK). The specifications also name Dashboard and Risk Radar. Controls as in Exhibit 14. ■ Source: Rosales at 2:35 and 4:31, for the apps and Copilot[9]; a training recording, for Explore[27]; Woodhead at 4:49, for Studio[1]; the public specifications and SDK[6, 29]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the Data Cloud launch release[34]; the FY2025 10-K, for hosting[12]; Microsoft, for Azure[35]; data vendors, ICE and Sustainalytics[36, 37].

Exhibit 4. What Studio is made of

WHAT STUDIO OFFERS

APIs
Read and write Aladdin data. Built on Aladdin Graph, BlackRock's firm-wide runtime for APIs and events. Every public specification's ID begins `agraph.<domain>`, for Aladdin Graph

Event-driven workflows
Aladdin tells a program when something happens. Events run on the same Aladdin Graph runtime. No event specification is public

Aladdin SDK
Open-source Python library with typed clients for the APIs and Data Cloud, with login, retries and rate limits built in. A BlackRock plugin builder generates its 11 plugin packages from the specifications

Aladdin Data Cloud
A managed data store on Snowflake, with Fivetran moving the data

Aladdin Compute
Hosted, ready-to-use Python environments

Aladdin Tools for Excel (ATX)
An Excel plugin that pulls Aladdin data into custom reports and analyses

Aladdin Data eXchange (ADX)
Bulk file transfer to and from Aladdin, with standard extracts and loaders

Developer Portal and Data Dictionary
Catalog, guides, field dictionary, notebooks and samples, behind a client login

Governance
"Data, tools, and governance support", including "access management and usage monitoring"

INSIDE A CLIENT'S STUDIO WORKSPACE

- **Studio UI** The screens where tokens, applications and storage are set up
- **API tokens and OAuth applications** Sign-in for the APIs
- **S3 storage integration** File storage for SDK code, set up in a Project
- **ADC Admin Center** The screen for managing Data Cloud datasets
- **Projects** Personal or shared, each holding OAuth applications and storage
- **Studio Compute sessions** Where Aladdin Compute runs SDK code
- **Studio notifications** Alerts sent through Studio's notification APIs

■ Source: BlackRock's Studio, APIs, ATX and ADX pages, which list Data Cloud, ADX, Aladdin Compute and ATX as "Additional Aladdin Studio resources"[14, 30, 32, 33]; the 54 public specifications[6], for the ids, the portal link and the Admin Center; the SDK README[29], for the SDK and the workspace; Naik and Kalish[38] and the Aladdin Graph job posting[39], for Aladdin Graph; the plugin builder[40]; Woodhead[1], Goel[41], the Aladdin Risk page[31] and a BlackRock job posting[42], for events and the portal; Fivetran[28]; Snowflake's partner page, for Compute[43]; the developer portal[44]; Studio for Aladdin Risk, for access management[45]. I read Studio Compute sessions as the SDK's name for Aladdin Compute.